

LIVE



NEWSPAPER

EDITORIALS

18th Apr, 2017

2/16th WEEK

1	HIV/AIDS Bill	00:00
2	Turkey... Presidential Republic	09:03
3	Jayalalithaa Abatement	13:49
4	Climate Change	23:15
5	Chemical Weapons	32:23

1

Legally enabling

The HIV/AIDS Bill provides a solid base for further empowerment and treatment access

The HIV and AIDS (Prevention and Control) Bill passed by Parliament does not guarantee access to anti-retroviral drugs and treatment for opportunistic infections, but there is no denying that it is a good base for an active health rights movement to build upon. Understandably, HIV-positive people in the country, estimated at over 21 lakh, are disappointed that the Centre's commitment to take all measures necessary to prevent the spread of HIV or AIDS is not reflected in the Bill, in the form of the right to treatment. The law only enjoins the States to provide access "as far as possible". Beyond this flaw, though, the legislation empowers those who have contracted the infection in a variety of ways: such as protecting against discrimination in employment, education, health-care services, getting insurance and renting property. It is now for the States to show strong political commitment, and appoint one or more ombudsmen to go into complaints of violations and submit reports as mandated by the law. Here again, State rules should prescribe a reasonable time limit for inquiries into complaints, something highlighted by the Standing Committee on Health and Family Welfare that scrutinised the legislation.

Access to insurance for persons with HIV is an important part of the Bill, and is best handled by the government. The numbers are not extraordinarily large and new cases are on the decline, according to the Health Ministry. Data for 2015 published by the Ministry show that two-thirds of HIV-positive cases are confined

THE  HINDU

Date : 18th April, 2017

AIDS ... FACTS & FIGURES



1 *India runs the second largest Antiretroviral Therapy programme in the world.*

2 *Govt. spent Rs. 2,000 cr on Antiretroviral Therapy alone and this is a 100% Centrally Sponsored Scheme.*

Parliament passed HIV and AIDS (Prevention and Control) Bill.

SALIENT POINTS

- It guaranteed equal rights in medical treatment, admission to educational institutions and jobs.
- This lists various grounds on which discrimination against HIV positive persons and those living with them is prohibited.
- Whoever does not adhere to the provisions of the bill will be penalised. There will be civil and criminal proceedings against them.
- The law provides a broad legislative framework with regard to the response to HIV in India.
- It is the first National HIV Law in South Asia.



FURTHER POINTS

- It applies to holding of public as well as private offices.
- Access to insurance and freedom of movement.
- It bans unfair treatment of people living with and affected by HIV with regard to accessing public facilities such as shops, restaurants, hotels, public entertainment venues, public facilities etc.

ANTIRETROVIRAL THERAPY (ART)



- 1** *It consist of combination of antiretroviral drugs to supress the HIV virus.*
- 2** *It also stops the progression of HIV disease.*
- 3** *It will result in huge reduction in deaths.*
- 4** *WHO recommends ART for all the people with HIV, as soon as possible after diagnosis without any restrictions of CD4 counts.*

WHAT IS CD4 COUNT?



CD4 cells are a type of white blood cells that play a major role in protecting your body from infection.

- 1** *CD4 cells are a type of white blood cells that play a major role in protecting body from infection.*
- 2** *The CD4 count is a lab test that measures the number of CD4 cells in a sample of blood.*
- 3** *Higher the CD4 count, the better able to fight HIV and other infections.*
- 4** *CD4 cells send signals to activate body's immune response, when they detect intruders like viruses or bacteria.*

WHAT IS CD4 COUNT?

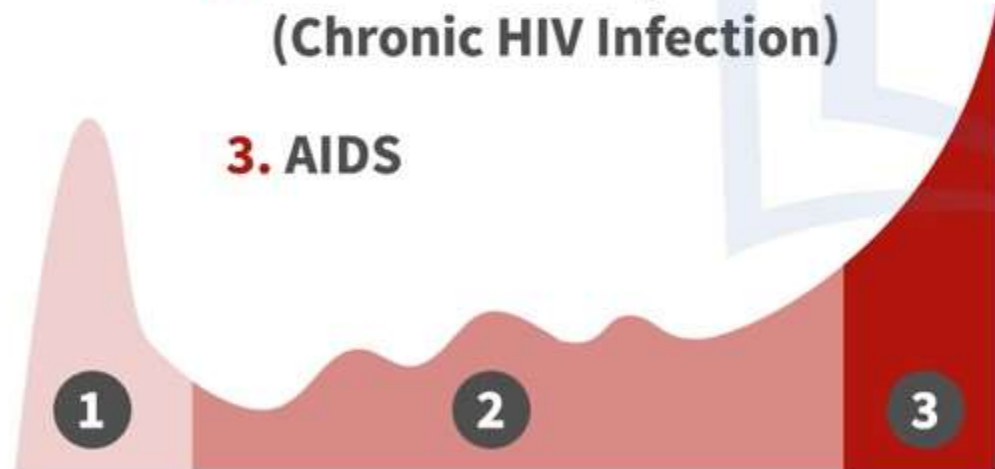
Left untreated, **HIV** progresses
in **3 STAGES:**

1. Acute infection
2. Clinical latency
(Chronic HIV Infection)
3. AIDS

5 When a person is infected with HIV, the virus begins to attack and destroy CD4 cells of the person's immune system.

6 CD4 count of uninfected adult is generally 500 cells per mm^3 to 1,600 cells per mm^3

7 A very low CD4 count of less than 200 cells per mm^3 is one of the ways to determine whether a person living with HIV has progressed to the stage 3 infection.



AIDS BILL ... CRITICAL APPRAISAL

- 1** *It doesn't guarantee access to antiretroviral drugs and treatment.*
- 2** *Hence, right to treatment is not incorporated.*
- 3** *The law basically enjoins states to provides access 'as far as possible'.*
- 4** *It basically protects against discrimination, education, employment and healthcare services.*

- 5** *Access to insurance for persons with HIV is important part.*
- 6** *AIDS Bill assumed significance in view of national commitment UN Sustainable Development Goals of which item no.3 is to end epidemic of AIDS by 2030.*

WHAT NEEDS TO BE DONE?

- 1** *Rapid scaling up of interventions to prevent new cases and offer free universal treatment is critical.*
- 2** *Publicly funded insurance can easily bring this subset of care seekers in overall risk pool.*
- 3** *Govt. should provide opportunities for education, skill building and employment.*
- 4** *Necessary guidelines are to be framed for the implementation of the law with a stringent framework in a reasonable time frame.*
- 5** *States must appoint one or more ombudsmen to go into complaints of violations.*
- 6** *State rules should prescribe reasonable time limit for enquiring into complaints.*

Periodic reports on compliance will exert pressure on states to meet their obligations.

2

Mr. Erdogan rising

The stage is set for Turkey's strongman to assume even more power

The path is now clear for Turkey to be transformed from a parliamentary democracy to a presidential republic, after a referendum on constitutional reforms proposed by the ruling Justice and Development Party (or AKP) gave the nod for handing sweeping powers to President Recep Tayyip Erdogan. The "Yes" campaign won by a relatively narrow margin, with a little more than 51% of the vote, and the opposition Republican People's Party (CHP) cited irregularities, including the use of unstamped ballot papers. The three biggest cities, Istanbul, Ankara and Izmir, voting "No" also indicates that much work remains to be done by the incumbents to bridge the rift within the polity. However, the head of the electoral body said the vote was valid. This remarkable turn of events, which will echo through the region and beyond, marks a step change from Turkey's historical tryst with representative democracy. The idea of major constitutional reforms of this sort has been in the making at least since 2014, when Mr. Erdogan became Turkey's first directly elected president. Nevertheless, many in Turkey and elsewhere, including anxious liberals across the EU, will watch with concern as the 18 major reforms on the table now will centralise power to an unprecedented extent in Mr. Erdogan's hands, raising valid questions about the separation of powers in the Turkish government.

The new executive powers that will accrue to Mr. Er-

THE  HINDU

Date : 18th April, 2017



ROMANIA

RUSSIA

Black Sea

Bosphorous Strait

GEORGIA

Ankara

Istanbul

TURKEY

Sea of Marmara

Dardanelles

Aegean Sea

Greece

ALBANIA

MACEDONIA (FYROM)

SERBIA

HUNGARY

CROATIA

MONTENEGRO

KOSOVO

AZERBA

SYRIA

IRAQ

Mediterranean Sea

Cyprus

Lebanon





CZECH REP

SLOVAKIA

UKRAINE

ANCE

SWITZERLAND

AUSTRIA

HUNGARY

MOLDOVA

SLOVENIA

CROATIA

Bosnia
and
Herzegovina

SERBIA

ROMANIA

RUSSI

Adriatic Sea

ITALY

KOSOVO

BULGARIA

Black Sea

MACEDONIA
(FYROM)

ALBANIA

Aegean
Sea

TURKEY

Greece

Ionian
Sea

Sea

GEOR

Balkan Peninsula

Mediterranean Sea

Cyprus

SYRIA

Lebanon

IRA

AT A GLANCE

- Official ... Republic of Turkey
- Major ... Anatolian Peninsula
- Minor ... Balkan Peninsula
- It is successor to the Ottoman Empire
- Republic of Turkey established on 29th Oct, 1923.
- First President... Kemal Ataturk
- Early member of NATO
- Capital ... Ankara
- Largest city ... Istanbul
- Official language ... Turkish
- Largest minority group ... Kurds
- Currency ... Turkish Lira
- HDI ... 0.761 (high)
- Per capita Nominal GDP ... \$ 9,562
- Predominantly Sunni Muslim



OTHER IMPORTANT ASPECTS

- 1** *Joining EU membership is its dream, But, several EU members objected to it. Moreover, it appears that Turkey is not interested at this juncture to join EU.*
- 2** *It is strategically located in continents of Europe and Asia and it exercises control over entry to Black Sea.*
- 3** *Stability in Turkey is very important, if one looks at overall stability in West Asia Region.*
- 4** *It is Parliamentary Republic, but now Referendum is held to convert it into Presidential Republic.*
- 5** *In recent times authoritarian President Recep Tayyip Erdogan propagating Islamist politics.*
- 6** *Turkish military, historically known for its Kemalist secularism is quite independent. But, this may change from now.*

2

Mr. Erdogan rising

The stage is set for Turkey's strongman to assume even more power

The path is now clear for Turkey to be transformed from a parliamentary democracy to a presidential republic, after a referendum on constitutional reforms proposed by the ruling Justice and Development Party (or AKP) gave the nod for handing sweeping powers to President Recep Tayyip Erdogan. The "Yes" campaign won by a relatively narrow margin, with a little more than 51% of the vote, and the opposition Republican People's Party (CHP) cited irregularities, including the use of unstamped ballot papers. The three biggest cities, Istanbul, Ankara and Izmir, voting "No" also indicates that much work remains to be done by the incumbents to bridge the rift within the polity. However, the head of the electoral body said the vote was valid. This remarkable turn of events, which will echo through the region and beyond, marks a step change from Turkey's historical tryst with representative democracy. The idea of major constitutional reforms of this sort has been in the making at least since 2014, when Mr. Erdogan became Turkey's first directly elected president. Nevertheless, many in Turkey and elsewhere, including anxious liberals across the EU, will watch with concern as the 18 major reforms on the table now will centralise power to an unprecedented extent in Mr. Erdogan's hands, raising valid questions about the separation of powers in the Turkish government.

The new executive powers that will accrue to Mr. Er-

NEWS REPORT

- 1 A referendum on constitutional reforms got 'Yes' vote with 51% of the votes polled.
- 2 This is for handing over sweeping powers to President Recep Tayyip Erdogan.
- 3 This marks a sharp change from Turkey's historical tryst with representative democracy.
- 4 This has been in the making at least since 2014, when Mr. Erdogan became Turkey's first directly elected President.

WHAT WILL HAPPEN WITH THIS 'YES' VOTE?

- 1** The country will be transformed from a Parliamentary Democracy to a Presidential Republic.
- 2** It will abolish the post of Prime Minister.
- 3** Transfer of that power to the President.
- 4** The President will be authority to appoint members to the Judiciary.
- 5** It removes the bar on the President maintaining party affiliation.
- 6** Then, overwhelming control by the political party over the state institutions may take place.
- 7** The judiciary and the security forces will be under the party's control.

Not only the problems stemming from the civil war in Syria, but also Turkey is going to face problems with the change towards Presidency with overwhelming powers.

3

Why the Jayalalithaa case matters

By dismissing Karnataka's review petition, the Supreme Court might have struck a blow against public interest



B.V. ACHARYA

The late Chief Minister J. Jayalalithaa's 20-year-old Disproportionate Assets (DA) case is no ordinary one. Its ramifications, legally, in the country are wide-ranging and severe. A case regarding acquisition of disproportionate assets by a public servant, under the Prevention of Corruption Act, stands on a slightly different footing from an ordinary criminal case. In the case of possessing disproportionate assets, the allegation is that a public servant amasses wealth by illegal means and the object of law is not merely to punish the offender but also to see that the offender or his/her legal representatives do not own or enjoy such illegally acquired assets.

The Chief Minister passed away on December 5, 2016. Orders in the DA case had been reserved six months prior to this, after all hearings had concluded on June 7, 2016. On February 14, the Supreme Court upheld the 'guilty' verdict of the Bengaluru trial court, sending the other three accused — V.K. Sasikala, J. Javarasi and V.N. Sudhakaram — to jail, with a penalty of ₹10 crore each. The first accused, Jayalalithaa, was no more and hence the court held that the charges against her had abated.

On March 21, the State of Karnataka filed a review petition challenging that part of the order which held that the case against Jayalalithaa had abated. Our argument was that when the death of the accused takes place long after

the arguments are concluded but before a judgment is pronounced, there will be no question of abatement of appeal.

But the Supreme Court, by dismissing on April 5 the review petition filed by the State of Karnataka, missed an opportunity to settle this issue. Consequentially, what the highest court of the country has done is to set a bad precedent in helping corrupt public servants.

Take the instance of an accused public servant choosing to commit suicide after acquiring huge property by illegal means. Legal representatives or heirs of the accused, according to the Supreme Court, can later enjoy the benefits of the illegally accrued wealth and property left behind, as the case against the accused public servant abates. This is a retrograde step in the march towards eradication of corruption in public life.

The question of abatement

Apart from the question as to whether a criminal appeal filed with leave under Article 136 of the Constitution of India will ever abate on the death of the accused, this particular case raised other equally important questions regarding alleged abatement where death has taken place after conclusion of the arguments and the judgment was reserved.

It is settled law that there is no hiatus (a break or a gap) between the date of conclusion of arguments and the date on which the judgment is ultimately delivered. A judgment is expected to be pronounced immediately after the conclusion of the arguments and pronouncing the judgment on a later date is only for the convenience of the court. Any event occurring between the date the judgment is reserved and the actual



date it was delivered on could not have any effect on the judgment which is ultimately pronounced.

Order XXII Rule 6 of the Code of Civil Procedure in unambiguous terms states that there will be no abatement of an appeal if the death is after judgment is reserved. It further clarifies that such judgment pronounced shall have the same force and effect as if the judgment was delivered on the date on which the arguments were concluded.

The Supreme Court itself has constitutionally applied this rule in quite a few civil appeals by holding that there is no abatement of appeal where the death is after the judgment was reserved. The Supreme Court rules also provide that in the case of an election petition, the proceedings will not abate on the death of a candidate if death is after judgment is reserved once arguments are concluded.

There is no principle or authority which can be pressed into service to hold that a different view is possible in the case of a criminal appeal. The Supreme Court, in clear terms, held that the provisions of the Code of Criminal Procedure are not applicable to the appeals filed before the Supreme Court, by applying for Special Leave under Article 136 of the Con-

stitution, though for the purpose of uniformity principles therein can be applied in suitable cases. The Supreme Court rules also do not provide for abatement of any criminal appeal. It can therefore be safely concluded that there is no constitutional or statutory provision providing for abatement of appeal, especially in a case where death has taken place after the judgment is reserved.

The abrupt conclusion of the Supreme Court that the appeal against Jayalalithaa has abated ignores the above said principle of law. It is also relevant to note that the case was never posted for further hearing after the death of the accused.

When judgment was pronounced on February 14, the court stated that the case against Jayalalithaa had abated, without any discussion on the questions involved. This finding was recorded without hearing the parties. Under the circumstances, it would have been appropriate for the Supreme Court to at least afford an opportunity to the parties to address arguments on this question and take a suitable decision. However, the court dismissed the review petition on merits, rejecting the request for oral hearing.

The legal implications arising out of the death of the accused after the judgment is reserved was not debated but the dismissal was recorded based on an erroneous view of law. The principle of *sub silentio* (action taken without notice, in legal terms) is thus applicable to the facts of the present case.

Reasons for review petition

In a section of the media an erroneous impression has been created that the State of Karnataka, in its greed to collect the fine amount of

₹100 crore imposed on Jayalalithaa by the trial court, has filed the review petition. The DA case was originally filed by the State of Tamil Nadu and Karnataka had to step into the case only after the direction of the Supreme Court, which transferred the case on a finding that the process of justice was being subverted in Tamil Nadu as the main accused held the post of Chief Minister of the State at the time.

The Supreme Court declared that the State of Karnataka is sole prosecuting agency in the case. It is only in obedience of the order of the Supreme Court that Karnataka has performed its role as sole prosecuting agency, so that there was a fair trial of the case. The State of Karnataka has no individual interest in the matter. The fine amount collected as also the confiscated assets could only benefit Tamil Nadu. Karnataka is not a beneficiary.

The right of the State of Karnataka is only for reimbursement of the expenses incurred in connection with the litigation (legal expenses) as ordered by the Supreme Court. Karnataka filed the review petition as it felt that an important question of law has been erroneously decided. It has chosen to do so only to fulfil its constitutional obligations. Now that the review petition has been dismissed, the case has ultimately reached its logical end. Karnataka can have the satisfaction of knowing that it has effectively performed the obligations imposed on it by the Supreme Court.

B.V. Acharya served as special public prosecutor and special counsel in the disproportionate assets case involving the late Tamil Nadu Chief Minister Jayalalithaa and AIADMK general secretary V.K. Sasikala.

THE HINDU

Date : 18th April, 2017



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

JAYALALITHAA CASE AND ABATEMENT

- 1 The case against J Jayalalithaa is 20 year old disproportionate assets case.
- 2 It is under the Prevention of Corruption Act.
- 3 This is when a public servant amasses wealth by illegal means.
- 4 It is not merely to punish the offender, but also to see that the offender or his/her legal representatives do not enjoy such illegally acquired assets.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

JAYALALITHAA CASE

- 1 Jayalalithaa passed away on December 5, 2016.
- 2 All the hearings had concluded on June 7, 2016.
- 3 On Feb 14, 2017 Supreme Court upheld the guilty verdict of Bengaluru trial court against three accused, leaving Jayalalithaa.
- 4 The court felt that Jayalalithaa was no more and Jayalalithaa's case stands abated.
- 5 On March 21, Karnataka filed a review petition against the abatement of Jayalalithaa conviction.
- 6 But Supreme Court also dismissed the April 5 review petition.



ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 1



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

It sets a bad precedent and the property illegally earned by the public representatives can be enjoyed by the legal heirs, if this becomes a precedent.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 2

- Let us take the case when public servant after amassing hundreds of crores illegally, chooses to commit suicide.
- If the same logic is applied, legal representatives or heirs of the accused can later enjoy the benefits of the illegally accrued wealth and property left behind.

Hence, it is a retrograde step towards the march for eradication of corruption from public life.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

- *Order XXII, Rule 6 of the code of Civil Procedure states that, there will be no abatement of an appeal, if the death is after the judgement is reserved.*

ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 3

- It is almost settled law that there is no hiatus (Break or gap) between the date of conclusion of arguments and the date on which the judgement is ultimately delivered.
- Normally judgement is expected to be delivered immediately after the conclusion of the arguments. Any event occurring between the date of judgement is reserved and the actual date it was delivered on should not have any effect on the judgement which is ultimately pronounced.
- This case violates this principle.



ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 4



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

The Supreme Court itself has constitutionally applied this rule in quite a few civil appeals, by stating that there is no abatement of appeal, where the death is after the judgement was reserved.



ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 5



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu

- As Supreme Court confirmed no abatement in such civil cases, there can be no principle or authority with a different view in case of criminal appeals.
- The Supreme Court rules also do not provide for abatement of any criminal appeal.

So, it can be concluded that there is no constitutional or statutory provision providing for abatement of appeal especially in a case where death has taken place after the judgement is reserved.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu



ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 6

- When the judgement was pronounced on February 14, the Supreme Court stated that the case against Jayalalithaa had abated without any discussion on the questions involved.
- No opportunity was given to the parties to address the argument on the question.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu



ISSUES REQUIRING DEEPER INSIGHT

ISSUE NO: 7

The legal implications arising out of the death of the accused after the judgement is reserved was not debated, but the dismissal was recorded based on an erroneous view of law.



Late J. Jayalalithaa
Former Chief Minister
Tamil Nadu



Karnataka by filing the review petition feels that important question of law has been erroneously decided.

4 The great climate churn

Regional and global planning is essential to combat extreme events



SUJATHA BYRAVAN

In recent months, unprecedented rates of glacier melts have been reported both in the Antarctic and the Arctic. "A massive crack in Antarctica's fourth biggest ice shelf has surged forward by at least 10 kilometres since early January," said *Nature* magazine in a recent article. Glaciers cover the terrain in both these regions, which have the only permanent ice sheets that still exist on earth today.

The earth has enjoyed a more or less stable temperature for the last 10,000 years. Prior to that there were several ice ages and periods of warmer temperature, also known as inter-glacials. The ice ages are believed to have been caused by small shifts in the earth's orbit, but all the reasons for the temperature fluctuations observed are not yet entirely understood.

About 5.3 to 2.6 million years ago, during the Pliocene epoch, global sea levels were close to 30 metres higher than they are today, while average global temperatures were only about three to four degrees Celsius warmer. What could happen in the current century, as a result of anthropogenic climatic change, remains a matter of great interest within the scientific and policymaking community.

The melting Antarctic

The Antarctic ice sheet is 14 million sq km in area and holds a large amount of frozen fresh water. (In comparison, the area of India's land mass is about 3.3 million sq km.) If all the ice over the Antarctic were to melt, sea levels would rise by about 60 metres. Parts of the ice sheet also flow into the ocean and do so through ice shelves that protrude into the water. Several media reports over the last few months have covered the expanding rift or crack along the Larsen C shelf in the Antarctic, which is expected to break off at any time. Larsen A and B collapsed in 1995 and 2002 respectively. Normally, ice shelves lose mass by the breaking off, or calving, of some of the portions and also by melting.



On this ice "If all the ice over the Antarctic were to melt, sea levels would rise by about 60 metres." Tourists in the Antarctic peninsula in 2010. >>>

When such large chunks break away from an ice shelf, they speed up the collapse of the entire shelf. Since this is attached to the rest of the glacier, these processes can increase the speed at which the glacier flows into the ocean. Thus, even though the Larsen C collapse by itself, since it is in the water, will not raise sea levels, it will hasten the melting of the glacier it is connected to.

In 2014, Eric Rignot, principal scientist for the Radar Science and Engineering Section at NASA's Jet Propulsion Laboratory, wrote that the retreat of ice in the Amundsen Sea sector of West Antarctica is inevitable, with major consequences for global sea levels. While the entire West Antarctic Ice Sheet (WAIS) may take a few hundred to a thousand years to completely melt, the process and the resultant collapse are now recognised as unstoppable.

Rising sea levels

In the Arctic, if all the ice in the Greenland ice sheet were to melt, it would raise global sea levels by about 7 metres (or 23 feet). For the last several years, glaciologists have noticed that ice melt in the summer has increased and covers a larger area than previous years. Scientists now realise that a lot of the recent melt has been due to increasing surface melt, in ad-

dition to calving or breaking off of chunks of ice.

Experts have known that there are feedback mechanisms that speed up glacier melt, exactly what these processes are and the rate by which they accelerate the melting remains an area of research. Soot and dust carried by air from various places, bacteria and algal pigments in the meltwater, any other pigments in the glacier can all reduce the reflection of the sunlight, thus increasing the absorption of heat energy by the ice. This consequently increases ice melt, which then absorbs more solar radiation, thus accelerating a feedback process. The meltwater flows into deep shafts, or moulines, that then speed up the flow of the glacier.

But there are also other phenomena that seem to have an influence on glacier melt. Temperatures in Northern Greenland have been much warmer and in fact, surface melt has doubled Greenland's contribution to sea level rise over the period 1992-2011 to 0.74 mm per year. Carbon dioxide concentrations have crossed 400 ppm in the atmosphere and are the highest they have been in the past 4,00,000 years.

Modelling glacier melt is very complex as it is affected by the temperature of the water, ocean currents and other factors still not entirely un-

derstood, along with various positive feedback mechanisms that can speed up the melting. The well-known climatologist James Hansen and his colleagues published a paper last year suggesting that sea level rise is a non-linear process and given what happened to sea levels in past geological periods, we should prepare for a rise of several metres over the next 50-150 years. This would imply that many of those alive today would likely see substantial increases in their lifetimes.

Global response needed

The global community is well aware that many large and densely populated cities are located along the coast and in low-lying deltas. Protecting the coast is an expensive undertaking and even then dikes, sea walls and similar structures provide only partial protection, based on studies undertaken by the Dutch Delta Committee and others. For India, the east coast, especially certain low-lying districts, are extremely vulnerable to intensive storms, which then lead to flooding, salt-water intrusion, and loss of land and livelihoods. On the west coast, while there are generally fewer storms, the concern is coastal erosion and flooding from sea level rise. The discussion regarding sea level rise and potential coastal impacts needs also to be understood not just as a coastal phenomenon, but also as an issue that ripples through the entire economy. Flooding in Chennai two years back did not affect just the land, but went through the economy as a whole and Swiss Re, the reinsurance company, has estimated losses to the economy due to the floods to be \$2.2 billion.

Then, enforcing the coastal regulation zone, protecting vulnerable districts and the most vulnerable communities which rely on ecosystems and the sea for their livelihoods are areas that need strengthening. Regional agreements related to refugees from climate effects need to be initiated. As a country which has generally been open to refugees from Tibet, Nepal, Afghanistan, Bangladesh and Sri Lanka, initiating and taking forward the conversation on regional planning for extreme events such as sea level rise would be important for India, the largest country in the region.

Sujatha Byravan is Principal Research Scientist at the Center for Study of Science, Technology & Policy, Bangalore

THE  HINDU

Date : 18th April, 2017

LOOK AT THE FOLLOWING FACTS!

- 1** *The Earth has enjoyed a more or less stable temperature for the last 10,000 years.*
- 2** *Prior to that, there were several ice ages and periods of warmer temperatures also known as interglacials.*
- 3** *About 5.3 to 2.6 million years ago, global sea levels were close to 30 mts higher than they are today.*
- 4** *The Antarctic ice sheet is 14 million sq kms in area and it holds large amount of frozen fresh water.*
- 5** *If all the ice over the Antarctic were to melt, sea levels would rise by about 60 mts.*
- 6** *In the Arctic, if all the ice in the Greenland ice sheet were to melt, it would rise global sea levels by about 7 mts.*



LOOK AT THE FOLLOWING FACTS!

8 *Temperatures in northern Greenland have been much warmer by 3 to 6°C above normal.*

9 *Surface melt has doubled Greenland's contribution to sea level rise over the period 1992 to 2011, to 0.74 mm per year.*

10 *In the last few months in Antarctica, cracks along the Larsen C shelf is widening and is expected to break off at any point of time.*

11 *Larsen A & B collapsed in 1995 and 2002 respectively.*

12 *Normally, ice shelves lose mass by breaking off or calving of some of the portions and also by melting.*



WHAT ARE THE REASONS?

- 1** *The concentration of carbon dioxide in the atmosphere has crossed 400 ppm and the highest they have been in the past 4 lakh years.*
- 2** *Soot and dust carried by air from various places, bacteria and algal pigments in the melt water and any other pigments in the glacier can reduce the reflection of sunlight, thus increasing the absorption of heat energy by the ice.*
- 3** *This process consequently increase ice melt and then absorbs more solar radiation thus accelerating a cyclical process.*
- 4** *The melt down flows into the deep shafts, then speed up the flow of the glacier.*
- 5** *Modelling glacier melt is very complex as it is affected by the temperatures of water, ocean currents and other factors and these are still not entirely understood.*

WHAT ARE THE CONSEQUENCES?

- 1** *Several large and densely populated cities are situated on the coastal and low lying deltas.*
- 2** *Substantial population reside on the coastal deltas and they will be badly affected.*
- 3** *India's east coast is extremely vulnerable to intensive storms which then leads to flooding, salt water intrusion, loss of land and livelihoods.*
- 4** *On the west coast, the concern is coastal erosion and flooding from the sea level rise.*



SUJATHA BYRAVAN

In recent months, unprecedented rates of glacier melts have been reported both in the Antarctic and the Arctic. "A massive crack in Antarctica's fourth-biggest ice shelf has surged forward by at least 10 kilometres since early January," said *Nature* magazine in a recent article. Glaciers cover the terrain in both these regions, which have the only permanent ice sheets that still exist on earth today.

The earth has enjoyed a more or less stable temperature for the last 10,000 years. Prior to that there were several ice ages and periods of warmer temperature, also known as inter-glacials. The ice ages are believed to have been caused by small shifts in the earth's orbit, but all the reasons for the temperature fluctuations observed are not yet entirely understood.

About 5.3 to 2.6 million years ago, during the Pliocene epoch, global sea levels were close to 30 metres higher than they are today, while average global temperatures were only about three to four degrees Celsius warmer. What could happen in the current century, as a result of anthropogenic climatic change, remains a matter of great interest within the scientific and policymaking community.

The melting Antarctic

The Antarctic ice sheet is 14 million sq km in area and holds a large amount of frozen fresh water. In comparison, the area of India's land mass is about 3.3 million sq km. If all the ice over the Antarctic were to melt, sea levels would rise by about 60 metres. Parts of the ice sheet also flow into the ocean and do so through ice shelves that protrude into the water. Several media reports over the last few months have covered the expanding rift or crack along the Larsen C shelf in the Antarctic, which is expected to break off at any time. Larsen A and B collapsed in 1995 and 2002 respectively. Normally, ice shelves lose mass by the breaking off, or calving, of some of the portions and also by melting.



On thin ice "If all the ice over the Antarctic were to melt, sea levels would rise by about 60 metres." Tourists in the Antarctic peninsula in 2010. *AP*

When such large chunks break away from an ice shelf, they speed up the collapse of the entire shelf. Since this is attached to the rest of the glacier, these processes can increase the speed at which the glacier flows into the ocean. Thus, even though the Larsen C collapse by itself, since it is in the water, will not raise sea levels, it will hasten the melting of the glacier it is connected to.

In 2014, Eric Rignot, principal scientist for the Radar Science and Engineering Section at NASA's Jet Propulsion Laboratory, wrote that the retreat of ice in the Amundsen Sea sector of West Antarctica is inevitable, with major consequences for global sea levels. While the entire West Antarctic Ice Sheet (WAIS) may take a few hundred to a thousand years to completely melt, the process and the resultant collapse are now recognised as unstoppable.

Rising sea levels

In the Arctic, if all the ice in the Greenland ice sheet were to melt, it would raise global sea levels by about 7 metres (or 23 feet). For the last several years, glaciologists have noticed that ice melt in the summer has increased and covers a larger area than previous years. Scientists now realise that a lot of the recent melt has been due to increasing surface melt, in ad-

dition to calving or breaking off of chunks of ice.

Experts have known that there are feedback mechanisms that speed up glacier melt, exactly what these processes are and the rate by which they accelerate the melting remains an area of research. Soot and dust carried by air from various places, bacteria and algal pigments in the meltwater, any other pigments in the glacier can all reduce the reflection of the sunlight, thus increasing the absorption of heat energy by the ice. This consequently increases ice melt, which then absorbs more solar radiation, thus accelerating a feedback process. The meltwater flows into deep shafts, or moulines, that then speed up the flow of the glacier.

But there are also other phenomena that seem to have an influence on glacier melt. Temperatures in Northern Greenland have been much warmer and in fact, surface melt has doubled Greenland's contribution to sea level rise over the period 1992-2011 to 0.74 mm per year. Carbon dioxide concentrations have crossed 400 ppm in the atmosphere and are the highest they have been in the past 4,00,000 years.

Modeling glacier melt is very complex as it is affected by the temperature of the water, ocean currents and other factors still not entirely un-

derstood, along with various positive feedback mechanisms that can speed up the melting. The well-known climatologist James Hansen and his colleagues published a paper last year suggesting that sea level rise is a non-linear process and given what happened to sea levels in past geological periods, we should prepare for a rise of several metres over the next 50-150 years. This would imply that many of those alive today would likely see substantial increases in their lifetimes.

Global response needed

The global community is well aware that many large and densely populated cities are located along the coast and in low-lying deltas. Protecting the coast is an expensive undertaking and even then dikes, sea walls and similar structures provide only partial protection, based on studies undertaken by the Dutch Delta Committee and others. For India, the east coast, especially certain low-lying districts, are extremely vulnerable to intensive storms, which then lead to flooding, salt water intrusion, and loss of land and livelihoods. On the west coast, while there are generally fewer storms, the concern is coastal erosion and flooding from sea level rise. The discussion regarding sea level rise and potential coastal impacts needs also to be understood not just as a coastal phenomenon, but also as an issue that ripples through the entire economy. Flooding in Chennai two years back did not affect just the land, but went through the economy as a whole and Saxis Re, the reinsurance company, has estimated losses to the economy due to the floods to be \$2.2 billion.

Then, enforcing the coastal regulation zone, protecting vulnerable districts and the most vulnerable communities which rely on ecosystems and the sea for their livelihoods are areas that need strengthening. Regional agreements related to refugees from climate effects need to be initiated. As a country which has generally been open to refugees from Tibet, Nepal, Afghanistan, Bangladesh and Sri Lanka, initiating and taking forward the conversation on regional planning for extreme events such as sea level rise would be important for India, the largest country in the region.

Sujatha Byravan is Principal Research Scientist at the Center for Study of Science, Technology & Policy, Bangalore

WHAT TO DO?

- 1 Enforcing Coastal Regulation Zone is the need of the hour.
- 2 Protecting vulnerable districts and most vulnerable communities which rely on the ecosystem and the sea for their livelihoods are the areas that need strengthening.
- 3 Regional agreements related to refugees from the climate affected areas needs to be initiated.

Regional planning for extreme events such as sea level rise would be important for India.

5

A lingering threat

There is fear that whatever remains of chemical stockpile may possibly have fallen into the hands of extremists

GARIMELLA SUBRAMANIAM



If confirmed by the Organisation for the Prohibition of Chemical Weapons (OPCW), the April 4 chemical attack in Syria, where more than 80 people suffocated to death, would mark a tragic coincidence of the 20-year commemoration of the ban on chemical arms.

While the incident is still under investigation by the OPCW, the bodies of victims are believed to bear signs of the effects of exposure to sarin gas, similar to the 2013 Ghouta attack.

When Syria confessed in 2012 to its chemical munitions stockpiles, assuring the world that it would not deploy them against non-combatants, it merely confirmed widely held suspicions. But the death of hundreds in Ghouta undermined then U.S. President Barack Obama's redline on Syria's use of chemical weapons. Under a subsequent deal that Moscow brokered with western nations, Washington, weary from the quagmire in Iraq, held back from punitive military action.

The horror in Ghouta came to be described as Syria's Halabja, a reference to the 1988 massacre of 5,000 Kurds in the Iraqi town. The global outrage against the 2013 incident forced Syrian President Bashar al-Assad to accede to the 1997 treaty.

The removal of the final consignment of chemicals in June 2014 to different destruction destinations marked a milestone in the OPCW's Syria operations. But questions have remained over the stockpiles and production facilities that Damascus has declared to international monitors. Within a year of the removal came the chlorine barrel bombing on Sarmin, in March 2015, killing six people. There have been similar instances before the horror that shook Khan Sheikhoun earlier this month.

THE  HINDU

Date : 18th April, 2017

WEAPONS OF MASS DESTRUCTION

CHEMICAL WEAPONS

- Mustard Gas
- Sarin
- Tabun
- VX Agent



Mustard Gas



Sarin



Tabun



VX Agent

TYPES OF CHEMICAL WEAPONS

1

BLISTER AGENTS

These oily substances contact with skin. They cause large scale skin blisters which resemble burns

- Sulphur Mustard
- Nitrogen Mustard

2

BLOOD AGENTS

They inhibit the ability of blood cells to utilise and transfer oxygen.

- Hydrogen Cyanide
- Cyanogen Chloride
- Arsine

3

NERVE AGENTS

Extremely Toxic and have rapid affect. Enters the body through inhalation or through skin.

- Tabun
- Sarin
- Soman
- VX

WHAT ABOUT CONVENTION ?



- Organisation for the Prohibition of Chemical Weapons (OPCW)
- Headquarters:
The Hague

- OPCW implements the “Chemical Weapons Convention”
- Goal is World “Free of Chemical Weapons”
- Entered into force in 1997
- It has got 192 member states

4 MAIN GOALS

- 1** Destroying all chemical weapons under International verification
- 2** Monitoring chemical industry to prevent re-emergence of new weapons
- 3** Protection to state parties against chemical threat
- 4** Fostering international cooperation for implementing the Convention

SALIENT POINTS

- 1** The Chemical Weapons Convention was concluded in 1997.
- 2** Subsequently, between 2004 and 2009 the U.S. soldiers detonated thousands of chemical weapons dangerously under open skies.
- 3** In 2009, Iraq acceded to the Chemical Weapons Convention.
- 4** In 2013, Syria used Chemical Weapons in Ghouta.
- 5** Subsequently, Syria acceded to the Chemical Weapons Convention.
- 6** Later on, the final consignment of chemicals were removed in June 2014 to the destruction destinations.
- 7** It is not clear how Syria is using these weapons. Within a year of the removal, Syria used these weapons on Sarmin and now on Khan Sheikhoun.

REPORT OF THE OPCW



ORGANISATION FOR THE
PROHIBITION OF CHEMICAL WEAPONS

1 As of 2016, 90% of the world's declared stockpile of chemical agents have been verifiably destroyed or converted to peaceful purposes.

2 Around 98% of the world is free from exposure to chemical arms.

The legal ban that has been in place for two decades is a significant advance, but the biggest worry is some of them might have gone into the hands of non-state actors like Islamic State.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao

D. Sunil Kumar
V. Dayakar James
Y. Sri Poojita

G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:
www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9